Methods in Python

22 June 2026 10:59

In Python a method is a function that belongs to a specific object or class. While a regular function is called independently by its name (like print()) a method is always invoked on object using dot notation (like Object.method()).

```
p = m.exp2(4)
q = m.exp(2)
print(f'e={m.e} to the power of 2 = {q}')
print(p)
```

Output:
```
e=2.718281828459045 to the power of 2 = 7.38905609893065
16.0
```

Method allows object to perform action on their own internal data, forming the backbone of object oriented programming (OOP).

In Python, a method is a function that belongs to an object or a class. Methods are used to perform operations on data associated with that object.

Difference between function and method

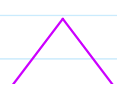
Function	**Method**
Independent block of code.	Function associated with an object or class.
Called directly.	Called through an Object
Example: print()	Example: math.exp()

Why use Methods?

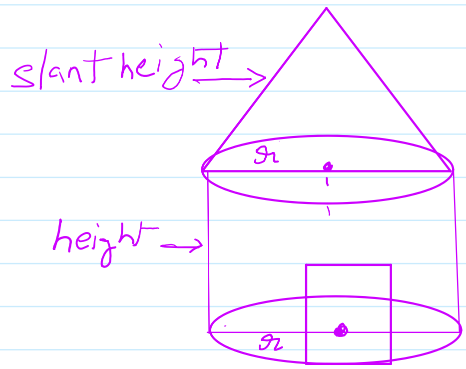
Method helps to:
- Organize code
- Reuse code
- Perform operations on objects
- Implement Object-Oriented Programming(OOP).

In programming, the use of function is one of the means to achieve modularity and reusability. Function can be defined as a named group of instructions that accomplish a specific task when it is invoked. Once defined, a function can be called repeatedly from different places of the program without writing all the code of the that function every time Or it can be called from inside another function by simply writing the name of the function and passing the required parameters if any. The programmer can define as many function as desired while writing the code.

slant height -



writing the code.




```python
# Program to calculate the cost of tent
#
import math
#  function definition
def areaOfCyl(h,r):
    area_cyl = 2*math.pi*r*h
    return area_cyl
# Function definition
def areaOfCone(l, r):
    area_con = math.pi*r*l #Area of conical part
    return area_con

#  function definition
def post_tax_price(cost):
    tax = 0.18 * cost
    net_price = cost + tax
    return net_price
#  main code
h = float(input("Height of cylinder: "))
r = float(input(" Enter radius of cylinder: "))
area_of_cylinder = areaOfCyl(h, r)
l = float(input(" Enter slant height of the conical area in meters : "))
concial_area = areaOfCone(l, r)
canvas_area = concial_area + area_of_cylinder
print(f"Area of canvas: {canvas_area:.2f} m^2")
#calculate the cost of canvas
unit_price = float(input(" Inter cost of 1m square Canvas in rupees: \u20b9 "))
total_cost = unit_price * canvas_area
print(f"The total cost of canvas before tax equal to \u20B9:{total_cost:.2f}")
print(f"Net amount payable(including tax) = \u20b9
{post_tax_price(total_cost):.2f}")
```

Output:

```
  Height of cylinder: 3
   Enter radius of cylinder: 1.5
   Enter slant height of the conical area in meters : 1.5
```

```
Area of canvas: 35.34 m^2
 Inter cost of 1m square Canvas in rupees: ₹ 50
The total cost of canvas before tax equal to ₹:1767.15
Net amount payable(including tax) = ₹ 2085.23
```